CHINMAY MHATRE

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SUMMARY

Computational chemist (Ph.D., Spring 2026) with demonstrated impact across four industry and government-sponsored programs. Builds and deploys multiscale simulation workflows spanning MLIP development, molecular dynamics, and DFT to translate atomistic findings into actionable materials design decisions. Two first-author publications in peer-reviewed journals.

EXPERIENCE

Research Associate - Computational Chemistry July 2026 – Sept 2026
University of Pittsburgh, Pittsburgh, Pennsylvania *Sponsor: Naval Research Laboratory*

- Engineered a high-throughput Python data pipeline to parse quantum chemistry (Gaussian) outputs and extract condensed-to-atoms density matrices, architecting a sharded HDF5 storage schema that accelerated I/O and downstream ML training across 100k+ QM9 structures and other related datasets.

Doctoral Researcher Aug 2020 – May 2026
Swanson School of Engineering, University of Pittsburgh, Pittsburgh, PA
Thesis: Atomistic modeling of Zr-based Metal-Organic Frameworks and stainless steel using classical and quantum mechanical methods

Machine Learning Interatomic Potentials for Reactive Systems *Funded by DTRA*

- Ran 100+ molecular dynamics simulations of catalytic kinetics on GPUs at extended timescales and identified surface poisoning as the root-cause failure mechanism, resolving catalytic performance loss unexplained by DFT at scale.
- Demonstrated non-oxidative carbon coupling on a stable transition-metal carbide stoichiometry using machine-learning potentials.
- Benchmarked an in-house active learning protocol against the published standard using statistical metrics, showing better accuracy at 50% fewer compute hours for the in-house protocol.
- Constructed a latent-space uncertainty metric to validate data novelty for configurations generated during active-learning protocol deployment.

Transport Modeling in Defective Porous Materials *Funded by DTRA*

- Modeled the physics of molecular transport through a defective nanoporous material with molecular dynamics (LAMMPS) and grand canonical Monte Carlo (RASPA), building a data-driven structure-property model linking defect configuration and concentration to diffusivity, and applied statistical analysis to extract diffusion coefficients across four concentrations and four defective structures.
- For a synthesizable defective MOF motif, showed a 100-fold transport improvement compared to the pristine MOF, enabled by defect engineering.
- Used collective variables (CV) and umbrella sampling to calculate the diffusivities of slow-moving toxic molecules in pristine UiO-66 using a classical potential, implementing dynamically corrected transition state theory.

Reaction Pathway & Electronic Structure Modeling *Funded by DTRA*

- Modeled the reaction energetics governing surface hydrolysis using first-principles DFT (VASP) and CI-NEB transition-state calculations, establishing the binding mechanism controlling reaction rates across four chemistries.

Reactive Force Field Model for Corrosion Physics *Funded by Naval Research Laboratory*

- Building a reactive force field model (Cr/O/Cl ReaxFF, LAMMPS) parametrized from first-principles data to capture the chemistry of chloride-induced pitting corrosion in oxide films at atomic resolution.

Thermophysical Property Model for Additive Screening *Funded by Lubrizol Corporation*

- Built a statistical-mechanical model (Gibbs ensemble Monte Carlo, Cassandra) to predict critical temperature and pressure for organometallic complexes with no experimental reference, validated liquid densities within 10% of available experimental data of these molecules through custom force field generation.

- Applied dimensionality reduction and data visualization to a conformational dataset to model the structure-property link between molecular geometry and dipole moment.

Graduate Student Intern May 2023 – Nov 2023
Chemical Separations Group, Oak Ridge National Laboratory, Oak Ridge, TN

- Built and benchmarked six free-energy solvation models against experimental data, using quantitative model-selection to deliver the validated best performer as the adopted DoE program standard.
- Applied data-driven structure-activity modeling across 10+ molecular modifications to rank candidates and accelerate the experimental design cycle.

SKILLS

Simulation	DFT, MD (FEP, metadynamics, MSD), MLIP development, GCMC, Gibbs ensemble MC, ReaxFF parametrization, vibrational frequency analysis
Software	LAMMPS, VASP, PLUMED, WHAM, PACKMOL, Gaussian, Orca, Quantum Espresso OpenMM, openbabel, CREST, pymol, RASPA
Machine Learning	PyTorch, TensorFlow, DeepMD (active learning), MACE, UMA, scikit-learn, CNN, GAN, VAE, equivariant neural networks, WandB, Google Cloud, AWS
Programming	Python (numpy, pandas, scipy, ASE, RDKit, UMAP, pymatgen, catkit,multiprocessing, Materials Project, OQMD), Linux/Bash, MATLAB, SQL, Git, HPC (Slurm, GPU), LaTeX
Data & Reports	Microsoft Office Suite, Power BI, DB Browser

PUBLICATIONS

Mhatre, C. V., Wardzala, J. J., Shukla, P. B., Agrawal, M., Johnson, J. K. (2023). Calculation of self, corrected, and transport diffusivities of isopropyl alcohol in UiO-66. *Nanomaterials*, 13(11), 1793. DOI:10.3390/nano13111793

Mhatre, C. V., Wardzala, J. J., Oliver, M. C., Islamov, M., Boone, P., Wilmer, C., Huang, L., Johnson, J. K. (2024). The Impact of Missing Linker Defects in UiO-66 on Adsorption and Diffusion of Isopropyl Alcohol. *The Journal of Physical Chemistry C*, 128(32), 13577–13587. DOI:10.1021/acs.jpcc.4c02982

Achar, S. K., Shukla, P. B., Mhatre, C. V., Bernasconi, L., Vinger, C. Y., Johnson, J. K. (2025). Reactive Active Learning: An Efficient Approach for Training Reaction Aware Machine Learning Potentials. *Journal of Chemical Theory and Computation*, 21, 18, 8889–8906. DOI:10.1021/acs.jctc.5c00920

Johnson, J. K., Kin Mai Lai, S., Mhatre, C. V., Williams, S., Keith, J. A., Call, M., Cormack, G. Estimation of Dipole Moments of Zinc Dialkyldithiophosphates and Their Precursors. (*under review*)

Mhatre, C. V., Gibbons, B., Morris, A. J., Johnson, K. Effect of Capping Group Binding on Chemical Reactivity.(*in preparation*)

Mhatre, C. V., Van Duin, A., Keith, J. A. Developing Cr/O/Cl ReaxFF Parameters to Understand Stainless Steel Depassivation in Saline Environments. (*in preparation*)

Mhatre, C. V., Gordon, A. Y., Ananthabhotala, L. Y., Achar, S., Johnson, J. K. Reactions of GB and HD in MOF-808-NH_2 Predicted from Machine Learning Interatomic Potentials. (*in preparation*)

EDUCATION

Ph.D., Chemical Engineering Aug 2020 - May 2026
University of Pittsburgh, Pittsburgh, USA
Advisors: Prof. J. Karl Johnson and Prof. John Keith

Bachelor of Chemical Engineering Aug 2016 – Aug 2020
Institute of Chemical Technology, Mumbai, India

SELECTED PRESENTATIONS (7 OF 11)

Achar, S., Mhatre, C., et al. (2025, Nov). Computational Design and ML Assisted Search of Dual Function MOFs Towards Chemical Warfare Agents' Decomposition. *AIChE Annual Meeting.*

Achar, S., Mhatre, C., et al. (2025, June). CH Bond Activation on Transition Metal Carbides: An MLFF Approach with Reactive Active Learning. *29th North American Catalysis Society Meeting.*

Mhatre, C., Policastro, S., Anderson, R., van Duin, A., Keith, J. (2024, Oct). Reactive Force Field Parametrizations to Examine Atomic Scale Processes on Stainless Steel in Chloride Environments. *AIChE Annual Meeting.*

Mhatre, C., Wardzala, J., et al. (2023, Nov). Diffusion and Adsorption of Polar Molecules in Defective UiO-66. *AIChE Annual Meeting.*

Mhatre, C., Gibbons, B., Morris, A. J., Johnson, K. (2023, Nov). Effect of Capping Group Binding on Chemical Reactivity. *AIChE Annual Meeting.*

Mhatre, C., Wardzala, J., et al. (2022, Nov). Modeling of Adsorption and Diffusion of Chemical Agents and Simulants in UiO-66. *AIChE Annual Meeting.*

Wardzala, J., Shukla, P. B., Mhatre, C., Johnson, K. (2021, Nov). Modeling of Diffusion in the Metal-Organic Framework UiO-66. *AIChE Annual Meeting.*

LEADERSHIP

Board Member, Engineering Graduate Student Organization (EGSO), University of Pittsburgh 2022 – 2025
Roles: Vice President, Secretary, Professional Development Officer
Coordinated graduate student events across the School of Engineering; managed a team of eight in planning and logistics for 500+ engineering graduates.

Teaching Assistant, University of Pittsburgh 2022 – 2024
Chemical Reaction Engineering, Chemical Engineering Thermodynamics, Mathematical Methods (avg. class size: 50). Developed course material and capstone projects.

AWARDS AND CERTIFICATIONS

- Pharmaceutical Formulations, Schrödinger Inc. 2026
- Introduction to Large Language Models, LinkedIn Learning 2025
- Deep Learning with PyTorch: Generative Adversarial Networks, Coursera 2025
- DTRA Grant: Computational design and ML-assisted search for dual-function MOFs towards nerve and blistering agent detoxification 2023
- ACS Reviewer Lab 2021
- Ongoing:
  - Databases and SQL for Data Science with Python, IBM/Coursera
  - Statistical Thinking for Industrial Problem Solving, JMP/Coursera
  - Building AI agents and Agentic Workflows, IBM/Coursera
  - Applied Bayesian Data Analytics, University of Pittsburgh/Coursera